

Fuzzy Reinforcement Learning-Based Safe Cooperative Control for Nonlinear Multiagent Systems

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Abstract—This article investigates safe cooperative control problem for nonlinear multiagent systems (MASs) with composite constraints and multiactuator faults. Specifically, the composite constraints primarily consist of state constraints and terminal time constraints. For the former, state-mapping functions and universal barrier functions are introduced to address asymmetric dynamic delayed constraints, which are further enhanced through state transformation to create new state variables. This approach eliminates the limitations imposed by initial system conditions and the influence of unknown terms, while accommodating both unconstrained and constrained scenarios. The latter introduces terminal time constraints on the basis of the former framework to rapidly satisfy system performance requirements. Furthermore, neural networks (NNs) are employed to approximate the reconstructed multiple fault information and system uncertainties. Then, an actor-critic-identifier architecture based on fuzzy reinforcement learning (RL) is constructed via an optimal backstepping (OB) method, enabling the solution of the Hamilton–Jacobi–Bellman equation within each subsystem without the need for persistent excitation. Finally, by integrating Lyapunov stability theory with graph theory, it is proven that all signals are bounded, and the followers ultimately achieve dynamic consensus with the leader. Simulation results are provided to demonstrate the effectiveness of this control strategy.

Index Terms—Dynamic delayed constraints, fuzzy reinforcement learning (RL), multiactuator faults, multiagent systems (MASs), optimal control.

I. INTRODUCTION

COOPERATIVE control of multiagent systems (MASs) has been proven to play a crucial role in industrial

applications, such as swarms of autonomous aerial vehicles [1], autonomous surface vehicles [2], [3], autonomous ground vehicles [4], [5], and mobile sensor networks [6]. Due to the influence of external complex environments and the internal operating mechanisms, the reliability and security of MASs need to always be guaranteed. Therefore, exploring safe control for MASs has consistently been a critical topic [7], [8], [9]. In recent years, constraint control and fault-tolerant control (FTC) have attracted much attention as an important branch of safe control, which aims to provide safe operating boundaries, reduce losses caused by actuator fault, and enhance the safe performance of MASs.

At present, experts and scholars have made relevant achievements in constraint control and FTC of MASs. Constraint control currently mainly involves state constraints, output constraints, input constraints, time constraints, and so on. Among these, state constraints are typically considered to meet safe performance requirements in actual MASs. The barrier Lyapunov function (BLF) is a fundamental tool for ensuring that state variables remain within prescribed boundaries [10], [11], [12], [13]. For example, in [10], a log-type BLF-based constraint controller was employed to ensure that all states remain within a predefined safe range for strict feedback MASs. In [11], a constraint controller was designed by combining an asymmetric log-type BLF with fuzzy logic system (FLS) for MASs, which avoids the need for feasibility conditions and partial derivative conditions. However, a key limitation of these approaches lie in its requirement for constrained variables to remain strictly confined within predefined bounds from the initial time, which imposes restrictions on the selection of constraint boundaries and may render the method ineffective when the initial values of the system states cannot meet these requirements. To address this problem, the concept of delayed state constraints was introduced in [14]. The core idea is to introduce a shift function that manages the initial values of tracking errors. Based on this idea, a delayed constraint controller for MASs was developed and employed to improve system safe [15], [16]. Although the above methods are effective in addressing state constraints, with the increasing scale and complexity of control systems, the likelihood of equipment failures becomes more significant. When an actuator fault occurs, system performance can be significantly degraded, potentially leading to even greater losses. Therefore, in addition to implementing state constraints, incorporating FTC mechanisms into MASs is crucial to ensure robust operation.

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FTC mainly categorizes into active and passive types, differing in their approach to fault-handling and response mechanisms [17]. Active FTC relies on fault detection and diagnosis, whereas passive FTC involves designing controllers that can tolerate certain faults without significant reliance on fault-specific information. Currently, most fault-handling methods involve the use of fault observers, adaptive control, and neural networks (NNs) to compensate for system losses [18], [19], [20], [21], [22], [23]. For example, a controller was developed for heterogeneous MASs, which solved the problems of multiple unknown dynamics directions and actuator faults [18], [19]. In [20], a fault observer was developed to evaluate existing fault problems to ensure the safe performance of MASs. In [21], [22], and [23], NN approximation methods were used to compensate for the unknown terms generated by fault control. Although the above methods have been proven to be very effective in handling actuator faults, most of them only consider a single fault type and do not consider the issue of state constraints. Moreover, few studies addressed the simultaneous handling of state constraint controls and actuator faults in unknown nonlinear MASs, making further exploration necessary and constituting one of the primary motivations for this study [24], [25]. In addition to the aforementioned considerations, it is essential to ensure the highest operational efficiency of MASs with minimal energy consumption in practical application scenarios. Therefore, it is necessary to further investigate the optimization problem to guarantee the optimal performance of MASs.

Solving the Hamilton–Jacobi–Bellman (HJB) equation is the key to achieving MAS’s optimal control. Generally, deriving an analytical solution to the HJB equation directly is extremely difficult since the presence of composite unknown terms. Moreover, the terminal time constraint acts as a boundary condition in optimal control, aiming to ensure that the system satisfies performance requirements within a finite time horizon while guaranteeing the boundedness of the performance function [26], [27], [28], [29]. Typically, the maximum principle or dynamic programming is employed to address terminal time constraints, but the computational difficulty increases with the time horizon. To address the above issues, adaptive dynamic programming (ADP) and reinforcement learning (RL) have been introduced [30], [31], [32]. A key advantage of these approaches is that ADP can adaptively update control parameters when the MASs are subjected to unknown dynamics or disturbances, and iterative learning ensures that the system achieves optimal performance. To this end, researchers have further extended the application of ADP and RL to constraint control or FTC [33], [34], [35]. For example, in [33], an optimal tracking strategy was proposed based on fault observer, which enhances the safe performance of unknown nonlinear interconnected systems. The distributed fault-tolerant consensus controller of MASs was developed by adopting ADP, and a local fault observer was established to ensure the safe of each agent [34]. In [35], an optimal backstepping (OB) technique was utilized to design a delayed state constraint controller by decomposing the optimal virtual controller and value function, effectively ensuring both local and global optimality. Despite these significant achievements,

most existing studies optimize safe control performance from a single perspective. Moreover, although some works have considered multiple aspects of optimal safe control, these are primarily limited to affine systems [36]. Therefore, they are an urgent need to develop an optimal strategy for high-order nonlinear MASs with safe performance, which simultaneously addresses both state constraints and multiactuator faults.

Inspired by the aforementioned results, this work proposes an optimal control strategy for unknown nonlinear MASs subject to state constraints and actuator faults. The main contributions are summarized as follows.

- 1) Different from previous works [24], [25], [36], which focused on either actuator faults or constraint issues, this article addresses the more challenging practical problem of composite constraints with actuator faults for unknown nonlinear MASs based on practical application. Specifically, the composite constraints include both state constraints and terminal time constraints, while the actuator faults encompass multiple types.
- 2) Distinct from the approaches in [12], [13], [15], [16], [26], [27], and [28], this work introduces dynamic delayed constraints to establish safe boundaries for MASs. This not only alleviates the dependence on initial states but also enhances flexibility under both constrained and unconstrained conditions. Furthermore, the use of terminal time constraints improves the rapid responsiveness of the MASs, avoids singularity issues while significantly reducing computational effort.
- 3) Unlike traditional methods that employ fault observer techniques to address unknown faults [18], [19], [20], [21], [22], [23], [24], [25], this article estimates the integrated nonlinear terms containing fault information and system uncertainties through system state reconstruction combined with NN-based approximation, effectively reducing practical design costs.

This article is organized as follows. Problem formulation and background are covered in Section II. Section III explores the system transformation, controller development, and stability proofs. Simulation results are presented in Section IV, followed by conclusion in Section V.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Graph Theory

The communication structure of MASs can be modeled using a directed graph $\mathcal{G} = (\mathcal{X}, \mathcal{O}, \mathcal{A})$. Here, $\mathcal{X} = \{1, 2, \dots, N\}$ and $\mathcal{O} = \{(i, j) : i, j \in \mathcal{X}\}$ are the sets of nodes and edges, respectively. $\mathcal{A} = [a_{i,j}] \in \mathbb{R}^{N \times N}$ is the corresponding adjacency matrix, where each element $a_{i,j}$ denotes the connectivity between agents, and it have $a_{i,j} = 1$ if $(i, j) \in \mathcal{O}$; otherwise, $a_{i,j} = 0$. Let $N_i = \{j \in \mathcal{X} : (i, j) \in \mathcal{O}\}$ denote the neighbors of agent i . The degree matrix $\mathcal{D} = \text{diag}(d_1, \dots, d_N)$ is defined such that $d_i = \sum_{j \in N_i} a_{i,j}$. The normalized adjacency matrix $\tilde{\mathcal{A}}$ is given by $\tilde{a}_{i,j} = a_{i,j}/d_i$ if $d_i \neq 0$ and $\tilde{a}_{i,j} = a_{i,j}$ otherwise. The normalization degree matrix $\tilde{\mathcal{D}}$ is defined as $\tilde{d}_i = 1$ if $d_i \neq 0$; otherwise, $\tilde{d}_i = 0$. The Laplacian matrix $\mathcal{L}$ can be obtained by $\mathcal{D} - \mathcal{A}$. The connectivity between the i th agent and the leader is

represented by the matrix $B = \text{diag}(b_1, b_2, \dots, b_N)$, where $b_i = 1$ if the i th agent receives information from the leader and $b_i = 0$ otherwise.

B. Fuzzy Logic Systems

An FLS based on fuzzy rules can be represented using IF-THEN statements.

$\mathcal{R}^q : \text{if } \xi_1 \in \mathcal{F}_1^q \wedge \xi_2 \in \mathcal{F}_2^q \wedge \dots \wedge \xi_n \in \mathcal{F}_n^q, \text{ THEN } f \in \mathcal{Y}^q, \quad q = 1, 2, \dots, m.$ Here, $\mathcal{F}_i^q$ and $\mathcal{Y}^q$ denote are fuzzy sets in $\mathbb{R}$, with $\xi = [\xi_1, \dots, \xi_n]^T$ being the input vector and f is fuzzy output. The output of the FLS is determined via the combination of crisp input fuzzification, multiplicative rule inference, and center-averaged defuzzification

$$f(\xi) = \frac{\sum_{q=1}^m \bar{f}_q \prod_{i=1}^n \mu_{\mathcal{F}_i^q}(\xi_i)}{\sum_{q=1}^m \prod_{i=1}^n \mu_{\mathcal{F}_i^q}(\xi_i)} \quad (1)$$

where $\mu_{\mathcal{F}_i^q}(\xi_i)$ is the membership function and $\bar{f}_q = \max_{f \in \mathcal{Y}^q} \mu_{\mathcal{Y}^q}(f)$ represents the centroid of the consequent membership function. Let the normalized firing strength be defined as follows:

$$S^q = \frac{\prod_{i=1}^n \mu_{\mathcal{F}_i^q}(\xi_i)}{\sum_{q=1}^m \left[\prod_{i=1}^n \mu_{\mathcal{F}_i^q}(\xi_i) \right]} \quad (2)$$

as the fuzzy activation function, and $W = [\bar{f}_1, \bar{f}_2, \dots, \bar{f}_m]^T$ and $S(\xi) = [S^1(\xi), S^2(\xi), \dots, S^m(\xi)]^T$. Therefore, the FLS is given by

$$f(\xi) = W^T S(\xi). \quad (3)$$

Lemma 1 ([40]): On the compact set Ω , $\varphi(\xi)$ is a continuous function. For $\varepsilon > 0$, there exists an FLS such that

$$\sup_{\xi \in \Omega} |\varphi(\xi) - W^T S(\xi)| < \varepsilon \quad (4)$$

where W denotes the fuzzy weight vector and ε is the fuzzy approximate error and is bounded.

C. System Description and Control Objective

An uncertain nonlinear MAS consisting of N followers and a leader agent. For each follower agent i th ($i = 1, 2, \dots, N$), the system is influenced by asymmetric dynamic delayed constraints and multiactuator faults, which is presented as follows:

$$\begin{cases} \dot{x}_{i,1} = g_{i,1}(\bar{x}_{i,1}) x_{i,2} + f_{i,1}(\bar{x}_{i,1}) \\ \dot{x}_{i,h} = g_{i,p}(\bar{x}_{i,h}) x_{i,h+1} + f_{i,p}(\bar{x}_{i,h}) \quad h = 2, \dots, n-1 \\ \dot{x}_{i,n} = g_{i,n}(\bar{x}_{i,n}) u_i^F + f_{i,n}(\bar{x}_{i,n}) \\ y_i = x_{i,1} \end{cases} \quad (5)$$

where $\bar{x}_{i,h} = [x_{i,1}, \dots, x_{i,h}]^T \in \mathbb{R}^h$ denotes the state variable of system, $f_{i,h}(\bar{x}_{i,h}) : \mathbb{R}^h \rightarrow \mathbb{R}$ is unknown continuous nonlinear function with $f_{i,h}(0) = 0$, and $g_{i,h}(\bar{x}_{i,h})$ represents the unknown control gain. The asymmetric dynamic delayed constraints refer to conditions where the state remains completely unconstrained over $t \in [0, T_d]$, and for $t \in [T_d, \infty)$, the state is effectively constrained within $(-\underline{\rho}_{i,h}, \bar{\rho}_{i,h})$, where $\underline{\rho}_{i,h}, \bar{\rho}_{i,h}$

are known and positive time-varying boundary functions. u_i^F is the system control input and may be subject to various types of actuator faults during system operation [37], [38], with the corresponding fault models described as follows:

$$u_i^F(t) = \begin{cases} u_{ia}(t) + \aleph_i(t), & \dot{\aleph}_i(t) \equiv 0, & \text{Bias} \\ \lambda_i(t) u_{ia}(t), & 0 < \lambda_i(t) \leq 1, & \text{Loss} \\ u_{ia}(t) + \aleph_i(t), & |\aleph_i(t)| = \mu t, 0 < \mu < 1, & \text{Drift.} \end{cases} \quad (6)$$

The actuator fault models can be transformed into a unified form

$$u_i^F(t) = \underbrace{\lambda_i(t) u_{ia}(t)}_{\text{(loss of effectiveness)}} + \underbrace{\aleph_i(t)}_{\text{(bias, drift)}} \quad (7)$$

where $u_{ia}(t)$ is the nominal control input, $\aleph_i(t)$ satisfying $|\aleph_i(t)| \leq \aleph_i(t)$ and $\lambda(t)$ represents the health indicator.

Accordingly, the leader system model is given by

$$\begin{cases} \dot{x}_0 = f_0(x_0) \\ y_0 = x_0 \end{cases} \quad (8)$$

where y_0 denotes the control output of leader and $f_0(x_0)$ is the known continuous nonlinear function.

The tracking error can be expressed as follows:

$$e_{i,1} = \sum_{j \in N_i} a_{i,j} (y_i - y_j) + b_i (y_i - y_0). \quad (9)$$

Control Objective: This article integrates safe considerations to develop an optimal control scheme for nonlinear MASs, achieving the following three objectives.

- 1) All system states are enforced to remain within the constraint boundaries for $t \in [0, T_d]$, i.e., $-\underline{\rho}_{i,h} < x_{i,h} < \bar{\rho}_{i,h}$ is satisfied for $\forall t \geq T_d$.
- 2) The closed-loop system ensures that all signals remain ultimately uniformly bounded.
- 3) Each follower agent's output y_i synchronizes with the leader's output y_0 .

III. OPTIMAL CONTROL DESIGN AND STABILITY ANALYSIS

In this section, to address the issues of asymmetric delayed state constraints and multiple actuator faults, we introduce several key components: a state-mapping function, a universal barrier function, an NN-based actuator fault compensator, and a coordinate transformation. Building on these elements, an optimal controller for the MASs is designed.

A. State-Mapping Transformation

To deal with delayed constraints [15], [16], [35], the state-mapping function is adopted as follows:

$$\mathcal{D}(t) = \begin{cases} 1 - \exp\left(1 - \frac{T_d}{T_d - t}\right), & 0 \leq t < T_d \\ 1, & t \geq T_d \end{cases} \quad (10)$$

where $T_d > 0$ is the time at which strict constraints on the state are imposed. Then, the auxiliary state variables for the follower and leader are constructed as follows:

$$x_{i,h}(t) = \mathcal{D}(t)x_{i,h}(t) + \mathcal{K}_{i,h}(1 - \mathcal{D}(t)), i = 0, 1, \dots, N \quad (11)$$

where $\mathcal{K}_{i,h} = (\bar{\rho}_{i,h} - \underline{\rho}_{i,h})/2$.

Remark 1: The state-mapping function $\mathcal{D}(t)$ satisfies the following three features: 1) $\mathcal{D}(t)$ is strictly monotonically increasing on $t \in [0, T_d]$ and remain invariable on $t \in [T_d, +\infty)$; 2) $\mathcal{D}(0) = 0$ and $\mathcal{D}(T_d) = 1$; and 3) $\mathcal{D}(t)$ is a continuously differentiable function. Furthermore, in most scenarios involving delayed constraints, the initial value of the auxiliary variable $x_{i,h}$ is typically constrained to zero, which is unsuitable when the boundary functions are $\bar{\rho}_{i,h} = 0$ or $\underline{\rho}_{i,h} = 0$. By contrast, using (11), the initial value of the auxiliary state variable $x_{i,h}(t)$ can be dynamically directed to the midpoint between the upper and lower boundaries, ensuring that the initial conditions are consistently satisfied.

B. Universal Barrier Function

Motivated by [41], in order to provide a safe operation boundary for the system, a universal barrier function is described as follows:

$$\mathcal{Z}_{i,h} = \frac{\bar{\rho}_{i,h}\underline{\rho}_{i,h}x_{i,h}}{(\bar{\rho}_{i,h} - x_{i,h})(\underline{\rho}_{i,h} + x_{i,h})}, \quad i = 0, 1, \dots, N. \quad (12)$$

Remark 2: The advantages of the $\mathcal{Z}_{i,h}$ function are primarily reflected in three aspects: 1) compared to log- and tan-type functions, the $\mathcal{Z}_{i,h}$ function is more flexible in handling asymmetric constraints, avoiding discontinuities caused by state switching. Moreover, $\mathcal{Z}_{i,h}$ can accommodate both symmetric and asymmetric constraints; 2) the initial condition $-\underline{\rho}_{i,h}(0) < x_{i,h}(0) < \bar{\rho}_{i,h}(0)$ can be automatically satisfied with $x_{i,h}(0) = (\bar{\rho}_{i,h}(0) - \underline{\rho}_{i,h}(0))/2$; and 3) when $\underline{\rho}_{i,h} = \bar{\rho}_{i,h} \rightarrow \infty$, we find that $\mathcal{Z}_{i,h} = x_{i,h}$. Thus, $x_{i,h}$ can effectively handle both constrained and unconstrained scenarios.

With the incorporation of state-mapping function (11) and (12), MASs are reconstructed as follows:

$$\dot{z}_{i,h} = x_{i,h} + \mathcal{Z}_{i,h}, i = 0, 1, \dots, N \quad (13)$$

further have

$$\begin{cases} \dot{z}_{i,1} = \Gamma_{i,1} + \Xi_{i,1} + \zeta_{i,2} \\ \dot{z}_{i,h} = \Gamma_{i,h} + \Xi_{i,h} + \zeta_{i,h+1}, & h = 2, \dots, n-1 \\ \dot{z}_{i,n} = \Gamma_{i,n} + \Xi_{i,n} + u_{ia} + \mathcal{F}_i \end{cases} \quad (14)$$

where $\Gamma_{i,h} = (\partial \zeta_{i,h} / \partial x_{i,h}) \dot{x}_{i,h} - \zeta_{i,h+1}$, $\Gamma_{i,n} = (\partial \zeta_{i,n} / \partial x_{i,n}) \dot{x}_{i,n} - u_{ia} - \mathcal{F}_i$, $\mathcal{F}_i = (\lambda_i - 1)u_{ia} + \mathfrak{N}_i$ and $\Xi_{i,h} = (\partial \zeta_{i,h} / \partial \mathcal{D}) \dot{\mathcal{D}} + (\partial \zeta_{i,h} / \partial \bar{\kappa}_{i,h}) \dot{\bar{\kappa}}_{i,h} + (\partial \zeta_{i,h} / \partial \underline{\rho}_{i,h}) \dot{\underline{\rho}}_{i,h}$, $\Xi_{i,n} = (\partial \zeta_{i,n} / \partial \mathcal{D}) \dot{\mathcal{D}} + (\partial \zeta_{i,n} / \partial \bar{\rho}_{i,n}) \dot{\bar{\rho}}_{i,n} + (\partial \zeta_{i,n} / \partial \underline{\rho}_{i,n}) \dot{\underline{\rho}}_{i,n}$, $i = 0, 1, \dots, N$.

On the basis of (14), the error (9) transforms into $z_{i,1} = \sum_{j \in N_i} a_{i,j}(\zeta_{i,1} - \zeta_{j,1}) + b_i(\zeta_{i,1} - \zeta_0)$, and it satisfies $|e_{i,1}| \leq |z_{i,1}|$.

Remark 3: Unlike traditional methods, this article introduces the state-mapping function (10), which ensures that the states $x_{i,h}$ converge into the desired constrained interval within a predefined time, while the transformed states $\zeta_{i,h}$ also satisfy the constraints. Additionally, it should be noted that a smaller predefined time T_d leads to faster convergence of the MASs, but also increases the computational burden. Therefore, when selecting T_d , a balance must be made between convergence speed and computational cost.

C. Coordinate Transformation

To facilitate the design of an optimal controller using the OB method, we need to perform a coordinate transformation based on (14)

$$\begin{cases} z_{i,1} = \sum_{j \in N_i} a_{i,j}(\zeta_{i,1} - \zeta_{j,1}) + b_i(\zeta_{i,1} - \zeta_0) \\ z_{i,h} = \zeta_{i,h} - \alpha_{i,h-1}, \quad h = 2, \dots, n \end{cases} \quad (15)$$

where $\alpha_{i,h}$ denotes the virtual controller. Additionally, to prevent the problem of differential explosion caused by the derivatives of the optimal virtual controller, dynamic surface control is introduced

$$\begin{aligned} \tau_{i,h} \dot{\alpha}_{if,h} &= \alpha_{i,h-1} - \alpha_{if,h} \\ \mathcal{B}_{i,h} &= \alpha_{if,h} - \alpha_{i,h-1} \end{aligned} \quad (16)$$

where $\alpha_{if,h}$ and $\mathcal{B}_{i,h-1}$ are the output and error of filter, respectively, and $\tau_{i,h} > 0$ is the filter adjustment parameter. Therefore, the performance index function of each agent is built as follows:

$$\mathcal{J}(z_{i,h}) = \int_0^{t_f} r_{i,h}(z_{i,h}(s), \alpha_{i,h}(z_{i,h})) ds \quad (17)$$

where $r_{i,h}(z_{i,h}, \alpha_{i,h}) = z_{i,h}^2 + \alpha_{i,h}^2$ is the utility function and it will be designed later and t_f stands for terminal time constraint.

D. Optimal Control Design

Before proceeding with the controller design, we need to establish some necessary knowledge preparation.

Assumption 1 ([15]): For each agent i , the function $g_{i,h}$ adheres to $0 < \underline{g}(\cdot) < |g_{i,h}| < \bar{g}(\cdot) < +\infty$, $i = 0, 1, \dots, N$, where $\underline{g}(\cdot) > 0$ and $\bar{g}(\cdot) > 0$.

Assumption 2 ([15], [35]): For the leader system y_0 , its trajectory is a continuously bounded time-varying function, and its h th derivative is also continuous and bounded.

Lemma 2 ([26], [27]): For the system $\dot{\eta} = f(\eta, t)$, practical finite-time stability is guaranteed if there exists a positive definite function $\mathcal{V}(\eta) : \Omega \rightarrow \mathbb{R}^+$ such that: 1) $\mathcal{V}(\eta) = 0$ if and only if $\eta = 0$ and 2) $\dot{\mathcal{V}}(\eta) \leq -\sigma \mathcal{V}^\kappa(\eta) + v$, where $\sigma > 0$, $v > 0$, and $0 < \kappa < 1$.

Definition 1 (Admissible Control [42]): For agent i , control input $u_i \in \Omega_u$ is classified as admissible under the following conditions: 1) $u_i(0) = 0$ and 2) the control input u_i ensures that both the consensus tracking error in the MASs and the associated performance function remain bounded, where Ω_u represents the set of all controls.

Lemma 3 ([39]): The following inequality holds for any $\psi \in \mathbb{R}$ $0 \leq |\psi| - \psi \tanh(\psi/\epsilon_*) \leq \varsigma_* \epsilon_*$, where $\epsilon_* > 0$ is a adjustable parameter and $\varsigma_* = 0.2785$.

Lemma 4 ([39]): The function $\Theta(\varpi) = \text{sig}^{2\varsigma-1}(\varpi) \tanh(|\varpi|^{2\varsigma}/\epsilon_*) : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, and its derivatives $\dot{\Theta}(\varpi), \ddot{\Theta}(\varpi)$ also keep continuity under the conditions $3/4 < \varsigma < 1, \epsilon_* > 0$, where $\text{sig}(\varpi) = \text{sign}(\varpi)|\varpi|^{2\varsigma-1}$.

Lemma 5 ([39]): For $\eta_1, \eta_2 \in \mathbb{R}$ and function $o(\eta_1, \eta_2) > 0$, there holds that $|\eta_1|^{t_1} |\eta_2|^{t_2} \leq (t_1 o|\eta_1|^{t_1+t_2} + t_2 o^{-t_1/t_2} |\eta_2|^{t_1+t_2}) / (t_1 + t_2)$.

Remark 4: In this article, Lemma 4 is introduced as an auxiliary function with continuous first- and second-order time derivatives, which is used to facilitate terminal time control. Its presence provides an alternative approach to address the singularity problems caused by repeated differentiation. In the process of stability analysis, Lemma 4 is applied together with Lemma 3, for example, in the case of “ $-z_{i,h} \Theta(z_{i,h}) = -|z_{i,h}|^{2\varsigma} \tanh(|z_{i,h}|^{2\varsigma}/\epsilon_*) \leq -(z_{i,h}^2)^{\varsigma} + \varsigma_* \epsilon_*$,” which is then directly used in the subsequent derivations.

The following is the procedure for designing the optimal safe controller.

Step 1: Based on (15), compute the time derivative of the error $z_{i,1}$ as follows:

$$\begin{aligned} \dot{z}_{i,1} &= \sum_{j \in N_i} a_{i,j} (\dot{z}_{i,1} - \dot{z}_{j,1}) + b_i (\dot{z}_{i,1} - \dot{z}_0) \\ &= \mathcal{E}_i (\Gamma_{i,1} + \Xi_{i,1} + \zeta_{i,2}) - b_i \dot{z}_0 - \Lambda_i \end{aligned} \quad (18)$$

where $\mathcal{E}_i = \sum_{j \in N_i} a_{i,j} + b_i$, $\Lambda_i = \sum_{j \in N_i} a_{i,j} \dot{z}_{j,1}$.

Let virtual controller $\alpha_{i,1} \triangleq \zeta_{i,2}$ further build the corresponding performance index function

$$\mathcal{J}_{i,1}(z_{i,1}) = \int_t^{t_f} r_{i,1}(z_{i,1}(s), \alpha_{i,1}(z_{i,1})) ds \quad (19)$$

where $\alpha_{i,1}^*$ and $\mathcal{J}_{i,1}^*$ are, respectively, defined as the optimality of the virtual controller and the performance function

$$\begin{aligned} \mathcal{J}_{i,1}^* &= \min_{\alpha_{i,1} \in \mathcal{M}_1} \left(\int_t^{t_f} r_{i,1}(z_{i,1}(s), \alpha_{i,1}(z_{i,1})) ds \right) \\ &= \int_t^{t_f} r_{i,1}(z_{i,1}(s), \alpha_{i,1}^*(z_{i,1})) ds \end{aligned} \quad (20)$$

where $\mathcal{M}_1$ represents the admissible strategy sets.

By computing (20), the HJB equation has

$$\begin{aligned} \mathcal{H}_{i,1}(z_{i,1}, \alpha_{i,1}^*, \nabla \mathcal{J}_{i,1}^*) &= r_{i,1}(z_{i,1}, \alpha_{i,1}^*) + \nabla \mathcal{J}_{i,1}^* \dot{z}_{i,1} \\ &= z_{i,1}^2 + \alpha_{i,1}^{*2} + \nabla \mathcal{J}_{i,1}^* \\ &\quad \times (\mathcal{E}_i (\alpha_{i,1}^* + \Gamma_{i,1} + \Xi_{i,1}) - b_i \dot{z}_0 - \Lambda_i) = 0. \end{aligned} \quad (21)$$

Then, the corresponding optimal controller is computed as follows:

$$\frac{\partial \mathcal{H}_{i,1}}{\partial \alpha_{i,1}^*} = 0 \Rightarrow \alpha_{i,1}^* = -\frac{1}{2} \mathcal{E}_i \nabla \mathcal{J}_{i,1}^*. \quad (22)$$

$\nabla \mathcal{J}_{i,1}^*$ is decomposed as follows:

$$\begin{aligned} \nabla \mathcal{J}_{i,1}^* &= \frac{2}{\mathcal{E}_i^2} \left(k_{i,1} \Theta(z_{i,1}) + \Upsilon_{i,1} + (1 + \mathcal{E}_i^2) z_{i,1} \right. \\ &\quad \left. + \mathcal{E}_i \Xi_{i,1} - b_i \dot{z}_0 + \nabla \bar{\mathcal{J}}_{i,1}^* \right) \end{aligned} \quad (23)$$

where $\nabla \bar{\mathcal{J}}_{i,1}^* = (-k_{i,1} \Theta(z_{i,1}) - \Upsilon_{i,1} - \mathcal{E}_i \Xi_{i,1} - (1 + \mathcal{E}_i^2) z_{i,1} + b_i \dot{z}_0 + (1/2) \mathcal{E}_i^2 \nabla \mathcal{J}_{i,1}^*) \in \mathbb{R}$ and $\Upsilon_{i,1} = \mathcal{E}_i \Gamma_{i,1} + \Lambda_i$ are unknown continuous function, $\Theta_{i,1}(z_{i,1}) = \text{sig}^{2\varsigma-1}(z_{i,1}) \tanh(|z_{i,1}|^{2\varsigma}/\epsilon_*)$. Correspondingly, the optimal virtual controller $\alpha_{i,1}^*$ is reconstructed as follows:

$$\begin{aligned} \alpha_{i,1}^* &= -\frac{1}{\mathcal{E}_i} \left(k_{i,1} \Theta(z_{i,1}) + \Upsilon_{i,1} + (1 + \mathcal{E}_i^2) z_{i,1} \right. \\ &\quad \left. + \mathcal{E}_i \Xi_{i,1} - b_i \dot{z}_0 + \nabla \bar{\mathcal{J}}_{i,1}^* \right). \end{aligned} \quad (24)$$

Since solving the unknown terms $\nabla \bar{\mathcal{J}}_{i,1}^*$ and $\Upsilon_{i,1}$ is extremely challenging, NNs are utilized to approximate them

$$\begin{aligned} \Upsilon_{i,1} &= W_{f_{i,1}}^{*\top} S_{f_{i,1}} + \varepsilon_{f_{i,1}} \\ \nabla \bar{\mathcal{J}}_{i,1}^* &= W_{\mathcal{J}_{i,1}}^{*\top} S_{\mathcal{J}_{i,1}} + \varepsilon_{\mathcal{J}_{i,1}} \end{aligned} \quad (25)$$

where $W_{f_{i,1}}^*$ and $W_{\mathcal{J}_{i,1}}^*$ denote the optimized weights, $S_{f_{i,1}}$ and $S_{\mathcal{J}_{i,1}}$ represent the activation functions, and $\varepsilon_{\mathcal{J}_{i,1}}$ and $\varepsilon_{f_{i,1}}$ represent the approximation deviations of NNs.

Substituting (25) into (23) and (24), we have

$$\begin{aligned} \nabla \mathcal{J}_{i,1}^* &= \frac{2}{\mathcal{E}_i^2} \left(k_{i,1} \Theta(z_{i,1}) + W_{f_{i,1}}^{*\top} S_{f_{i,1}} + \mathcal{E}_i \Xi_{i,1} \right. \\ &\quad \left. + (1 + \mathcal{E}_i^2) z_{i,1} - b_i \dot{z}_0 + W_{\mathcal{J}_{i,1}}^{*\top} S_{\mathcal{J}_{i,1}} + \varepsilon_{i,1} \right) \\ \alpha_{i,1}^* &= -\frac{1}{\mathcal{E}_i} \left(k_{i,1} \Theta(z_{i,1}) + W_{f_{i,1}}^{*\top} S_{f_{i,1}} + \mathcal{E}_i \Xi_{i,1} \right. \\ &\quad \left. + (1 + \mathcal{E}_i^2) z_{i,1} - b_i \dot{z}_0 + W_{\mathcal{J}_{i,1}}^{*\top} S_{\mathcal{J}_{i,1}} + \varepsilon_{i,1} \right) \end{aligned} \quad (26)$$

where $\varepsilon_{i,1} = \varepsilon_{f_{i,1}} + \varepsilon_{\mathcal{J}_{i,1}}$. $W_{f_{i,1}}^*$ and $W_{\mathcal{J}_{i,1}}^*$ as the ideal weights, and we need to introduce local identifier NN, actor NN, and critic NN to approximate them.

First, the unknown term $\Upsilon_{i,1}$ needs to be processed by using an identifier NN. Then, $\hat{\Upsilon}_{i,1}$ and $\hat{W}_{f_{i,1}}$ are described as follows:

$$\begin{aligned} \hat{\Upsilon}_{i,1} &= \hat{W}_{f_{i,1}}^\top S_{f_{i,1}} \\ \dot{\hat{W}}_{f_{i,1}} &= \gamma_{f_{i,1}} \left(S_{f_{i,1}} z_{i,1} - \sigma_{i,1} \hat{W}_{f_{i,1}} \right) \end{aligned} \quad (27)$$

where $\hat{W}_{f_{i,1}}$ is the weight estimate of $W_{f_{i,1}}^*$, $\hat{\Upsilon}_{i,1}$ is the output of identifier NN, $\gamma_{f_{i,1}} > 0$ is the gain matrix, and $\sigma_{i,1} > 0$ is a adjustment parameter.

Second, a local critic NN is utilized to evaluate unknown function $\nabla \bar{\mathcal{J}}_{i,1}^*$; then, $\nabla \hat{\mathcal{J}}_{c_{i,1}}$ and $\hat{W}_{c_{i,1}}$ are formulated as follows:

$$\begin{aligned} \nabla \bar{\mathcal{J}}_{i,1}^* &\Leftarrow \nabla \hat{\mathcal{J}}_{c_{i,1}} = \hat{W}_{c_{i,1}}^\top S_{\mathcal{J}_{i,1}} \\ \dot{\hat{W}}_{c_{i,1}} &= -\gamma_{c_{i,1}} S_{\mathcal{J}_{i,1}} S_{\mathcal{J}_{i,1}}^\top \hat{W}_{c_{i,1}} \end{aligned} \quad (28)$$

where $\hat{W}_{c_{i,1}}$ is the weight estimate of $W_{\mathcal{J}_{i,1}}^*$, $\nabla \hat{\mathcal{J}}_{c_{i,1}}$ stand for the output of local critic NN, $\gamma_{c_{i,1}} > 0$ is learning rate.

Next, the performance of $\nabla \bar{\mathcal{J}}_{i,1}^*$ is improved by using a local actor NN. Then, $\nabla \hat{\mathcal{J}}_{a_{i,1}}$ and $\hat{W}_{a_{i,1}}$ are built as follows:

$$\begin{aligned} \nabla \bar{\mathcal{J}}_{i,1}^* &\Leftarrow \nabla \hat{\mathcal{J}}_{a_{i,1}} = \hat{W}_{a_{i,1}}^\top S_{\mathcal{J}_{i,1}} \\ \dot{\hat{W}}_{a_{i,1}} &= \dot{\hat{W}}_{c_{i,1}} - \gamma_{a_{i,1}} S_{\mathcal{J}_{i,1}} S_{\mathcal{J}_{i,1}}^\top (\hat{W}_{a_{i,1}} - \hat{W}_{c_{i,1}}) \end{aligned} \quad (29)$$

where $\nabla \hat{\mathcal{J}}_{a,i,1}$ and $\hat{W}_{c,i,1}$ stand for the output of the local actor NN and the estimation of weights, respectively, and $\gamma_{a,i,1} > 0$ is the learning rate.

Finally, we successfully constructed a new sub-actor-critic-identifier framework as follows:

$$\begin{aligned} \nabla \hat{\mathcal{J}}_{i,1}^* &= \frac{2}{\varepsilon_i^2} \left(k_{i,1} \Theta(z_{i,1}) + \hat{W}_{f,i,1}^\top S_{f,i,1} + \mathcal{E}_i \Xi_{i,1} \right. \\ &\quad \left. + (1 + \varepsilon_i^2) z_{i,1} - b_i \dot{z}_0 + \hat{W}_{c,i,1}^\top S_{\mathcal{J},i,1} \right) \\ \hat{\alpha}_{i,1}^* &= -\frac{1}{\varepsilon_i} \left(k_{i,1} \Theta(z_{i,1}) + \hat{W}_{f,i,1}^\top S_{f,i,1} + \mathcal{E}_i \Xi_{i,1} \right. \\ &\quad \left. + (1 + \varepsilon_i^2) z_{i,1} - b_i \dot{z}_0 + \hat{W}_{a,i,1}^\top S_{\mathcal{J},i,1} \right). \end{aligned} \quad (30)$$

To illustrate the updating process of NN, substituting (30) into the HJB (13), we can get

$$\begin{aligned} \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*) &= z_{i,1}^2 + \hat{\alpha}_{i,1}^{*2} + \nabla \hat{\mathcal{J}}_{i,1}^* \\ &\quad \times (\mathcal{E}_i \hat{\alpha}_{i,1}^* + \Upsilon_{i,1} + \mathcal{E}_i \Xi_{i,1} - b_i \dot{z}_0). \end{aligned} \quad (31)$$

In sequence, $\mathcal{H}_{i,1}$ as the Bellman residual error, it has

$$\begin{aligned} \mathcal{H}_{i,1} &= \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*) - \mathcal{H}_{i,1}(z_{i,1}, \alpha_{i,1}^*, \nabla \mathcal{J}_{i,1}^*) \\ &= \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*). \end{aligned} \quad (32)$$

It is evident that optimized controller $\hat{\alpha}_{i,1}^*$ needs to guarantee $\mathcal{H}_{i,1} = \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*) \rightarrow 0$. When $\hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*) = 0$ is satisfied, its solution is unique

$$\begin{aligned} \frac{\partial \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*)}{\partial \hat{W}_{a,i,1}} &= \frac{2}{\varepsilon_i^2 \Gamma_{i,1}^2} S_{\mathcal{J},i,1} S_{\mathcal{J},i,1}^\top (\hat{W}_{a,i,1} - \hat{W}_{c,i,1}) = 0. \end{aligned} \quad (33)$$

To ensure that the learning update rate under the sub-actor-critic-identifier framework satisfies (33), we define the following positive definite function as follows:

$$\mathcal{Q}_{i,1} = (\hat{W}_{a,i,1} - \hat{W}_{c,i,1})^\top (\hat{W}_{a,i,1} - \hat{W}_{c,i,1}). \quad (34)$$

Then, the update laws (28) and (29) are derived through the application of gradient descent optimization.

Noting that $\partial \mathcal{Q}_{i,1} / \partial \hat{W}_{a,i,1} = 2\hat{W}_{a,i,1} - 2\hat{W}_{c,i,1} = -\partial \mathcal{Q}_{i,1} / \partial \hat{W}_{c,i,1}$, $\dot{\mathcal{Q}}_{i,1}$ is calculated as follows:

$$\begin{aligned} \dot{\mathcal{Q}}_{i,1} &= \frac{\partial \mathcal{Q}_{i,1}}{\partial \hat{W}_{a,i,1}} \dot{\hat{W}}_{a,i,1} + \frac{\partial \mathcal{Q}_{i,1}}{\partial \hat{W}_{c,i,1}} \dot{\hat{W}}_{c,i,1} \\ &= -\frac{\gamma_{a,i,1}}{2} \frac{\partial \mathcal{Q}_{i,1}}{\partial \hat{W}_{a,i,1}} S_{\mathcal{J},i,1} S_{\mathcal{J},i,1}^\top \frac{\partial \mathcal{Q}_{i,1}}{\partial \hat{W}_{a,i,1}} \leq 0. \end{aligned} \quad (35)$$

This implies that (33) can be achieved through the update rates (28) and (29).

Remark 5: Due to the inherent nonlinearity of the HJB (21), the optimal performance index function $\nabla \mathcal{J}_{i,1}^*$ and the corresponding optimal virtual controller $\alpha_{i,1}^*$ are generally unknown and cannot be solved analytically. To address this issue, a local critic NN and a local actor NN are employed to approximate

the ideal function $\nabla \bar{\mathcal{J}}_{i,1}^*$, enabling the construction of the estimated sub-actor $\hat{\alpha}_{i,1}^*$ and sub-critic $\nabla \hat{\mathcal{J}}_{i,1}^*$. However, since these approximations deviate from the true optimal solutions, substituting them into the HJB (21) results in an approximation error, denoted as $\mathcal{H}_{i,1}$. This error guides the online adaptation of the NN weights $\hat{W}_{a,i,1}$ and $\hat{W}_{c,i,1}$ through a simplified positive definite function, thereby enabling the system to achieve optimal control performance.

Remark 6: Under the sub-actor-critic-identifier network architecture, the critic and actor NNs are used to evaluate and improve the performance of the virtual controller $\hat{\alpha}_{i,1}^*$. Through iterative learning, the optimal solution $\hat{\alpha}_{i,1}^* \rightarrow \alpha_{i,1}^*$ and Bellman error $\mathcal{H}_{i,1} = \hat{\mathcal{H}}_{i,1}(z_{i,1}, \hat{\alpha}_{i,1}^*, \nabla \hat{\mathcal{J}}_{i,1}^*) \rightarrow 0$ are satisfied. Compared with the traditional ADP algorithm [30], [31], [32], the gradient update is performed based on a simple positive function, which significantly simplifies the adaptive rules in optimal control algorithms and eliminates the requirement for the persistent excitation condition. The necessity description has been given in (33)–(35).

Step h: In light of the error variable $z_{i,h} = \zeta_{i,h} - \alpha_{if,h}$, we can obtain

$$\dot{z}_{i,h} = \zeta_{i,h+1} + \Gamma_{i,h} + \Xi_{i,h} - \dot{\alpha}_{if,h}. \quad (36)$$

Define virtual controller $\alpha_{i,h} \triangleq \xi_{i,h+1}$, and the performance index function with respect to $z_{i,h}$ is given by

$$\mathcal{J}_{i,h}(z_{i,h}) = \int_t^{t_f} r_{i,h}(z_{i,h}(s), \alpha_{i,h}(z_{i,h})) ds. \quad (37)$$

The performance index function incorporating the optimal virtual controller $\alpha_{i,h}^*$ is formulated as follows:

$$\begin{aligned} \mathcal{J}_{i,h}^* &= \min_{\alpha_{i,h} \in \mathcal{M}_h} \left(\int_t^{t_f} r_{i,h}(z_{i,h}(s), \alpha_{i,h}(z_{i,h})) ds \right) \\ &= \int_t^{t_f} r_{i,h}(z_{i,h}(s), \alpha_{i,h}^*(z_{i,h})) ds \end{aligned} \quad (38)$$

where $\mathcal{M}_h$ represents the admissible strategy sets.

Then, the HJB equation is derived as follows:

$$\begin{aligned} \mathcal{H}_{i,h}(z_{i,h}, \alpha_{i,h}^*, \nabla \mathcal{J}_{i,h}^*) &= r_{i,h}(z_{i,h}, \alpha_{i,h}^*) + \nabla \mathcal{J}_{i,h}^* \dot{z}_{i,h} \\ &= z_{i,h}^2 + \alpha_{i,h}^{*2} + \nabla \mathcal{J}_{i,h}^* \\ &\quad \times (\alpha_{i,h}^* + \Gamma_{i,h} + \Xi_{i,h} - \dot{\alpha}_{if,h}) = 0. \end{aligned} \quad (39)$$

Thus, the corresponding optimal controller is

$$\frac{\partial \mathcal{H}_{i,h}}{\partial \alpha_{i,h}^*} = 0 \Rightarrow \alpha_{i,h}^* = -\frac{1}{2} \nabla \mathcal{J}_{i,h}^*. \quad (40)$$

Similar to *Step 1*, the partial differential function $\nabla \mathcal{J}_{i,h}^*$ and the optimal virtual control $\alpha_{i,h}^*$ are established as follows:

$$\begin{aligned} \nabla \mathcal{J}_{i,h}^* &= 2 \left(k_{i,h} \Theta(z_{i,h}) + \Upsilon_{i,h} + \frac{5}{2} z_{i,h} \right. \\ &\quad \left. + \Xi_{i,h} - \dot{\alpha}_{if,h} + \nabla \bar{\mathcal{J}}_{i,h}^* \right) \\ \alpha_{i,h}^* &= - \left(k_{i,h} \Theta(z_{i,h}) + \Upsilon_{i,h} + \frac{5}{2} z_{i,h} \right. \\ &\quad \left. + \Xi_{i,h} - \dot{\alpha}_{if,h} + \nabla \bar{\mathcal{J}}_{i,h}^* \right) \end{aligned} \quad (41)$$

where $\nabla \tilde{\mathcal{J}}_{i,h}^* = (-k_{i,h}\Theta(z_{i,h}) - \Upsilon_{i,h} - (5/2)z_{i,h} - \Xi_{i,h} - \dot{\alpha}_{if,h} + (1/2)\nabla \mathcal{J}_{i,h}^*) \in \mathbb{R}$ and $\Upsilon_{i,h} = \Gamma_{i,h}$ are unknown continuous function, $\Theta_{i,h}(z_{i,h}) = \text{sig}^{2\zeta-1}(z_{i,h}) \tanh(|z_{i,h}|^{2\zeta}/\epsilon_*)$.

By using the following NNs to approximate the unknown terms $\nabla \tilde{\mathcal{J}}_{i,h}^*$ and $\Upsilon_{i,h}$:

$$\begin{aligned}\Upsilon_{i,h} &= W_{f_{i,h}}^{*\top} S_{f_{i,h}} + \varepsilon_{f_{i,h}} \\ \nabla \tilde{\mathcal{J}}_{i,h}^* &= W_{\mathcal{J}_{i,h}}^{*\top} S_{\mathcal{J}_{i,h}} + \varepsilon_{\mathcal{J}_{i,h}}\end{aligned}\quad (42)$$

where $W_{f_{i,h}}^*$ and $W_{\mathcal{J}_{i,h}}^*$ are the ideal weights, $S_{f_{i,h}}$ and $S_{\mathcal{J}_{i,h}}$ denote the activation functions, and $\varepsilon_{\mathcal{J}_{i,h}}$ and $\varepsilon_{f_{i,h}}$ represent the approximation errors.

By substituting (42) into (41), we can reformulate the expression as follows:

$$\begin{aligned}\nabla \mathcal{J}_{i,h}^* &= 2 \left(k_{i,h}\Theta(z_{i,h}) + W_{f_{i,h}}^{*\top} S_{f_{i,h}} + \Xi_{i,h} + \frac{5}{2}z_{i,h} \right. \\ &\quad \left. - \dot{\alpha}_{if,h} + W_{\mathcal{J}_{i,h}}^{*\top} S_{\mathcal{J}_{i,h}} + \varepsilon_{i,h} \right) \\ \alpha_{i,h}^* &= - \left(k_{i,h}\Theta(z_{i,h}) + W_{f_{i,h}}^{*\top} S_{f_{i,h}} + \Xi_{i,h} + \frac{5}{2}z_{i,h} \right. \\ &\quad \left. - \dot{\alpha}_{if,h} + W_{\mathcal{J}_{i,h}}^{*\top} S_{\mathcal{J}_{i,h}} + \varepsilon_{i,h} \right)\end{aligned}\quad (43)$$

where $\varepsilon_{i,h} = \varepsilon_{f_{i,h}} + \varepsilon_{\mathcal{J}_{i,h}}$.

Similarly, we directly use a local actor-critic-identifier NN to approximate them

$$\begin{aligned}\Upsilon_{i,h} &\Leftarrow \hat{\Upsilon}_{i,h} = \hat{W}_{f_{i,h}}^\top S_{f_{i,h}} \\ \nabla \tilde{\mathcal{J}}_{i,h}^* &\Leftarrow \nabla \hat{\mathcal{J}}_{i,h} = \hat{W}_{c_{i,h}}^\top S_{\mathcal{J}_{i,h}} \\ \nabla \tilde{\mathcal{J}}_{i,h}^* &\Leftarrow \nabla \hat{\mathcal{J}}_{ai,h} = \hat{W}_{ai,h}^\top S_{\mathcal{J}_{i,h}}\end{aligned}\quad (44)$$

where $\nabla \hat{\mathcal{J}}_{ci,h}^*$, and $\nabla \hat{\mathcal{J}}_{ai,h}^*$ and $\hat{\Upsilon}_{i,h}$ are the estimates of $\nabla \mathcal{J}_{i,h}^*$ and $\Upsilon_{i,h}$, respectively. The update rates of $\hat{W}_{f_{i,h}}$, $\hat{W}_{c_{i,h}}$, and $\hat{W}_{ai,h}$ have

$$\begin{aligned}\dot{\hat{W}}_{f_{i,h}} &= \gamma_{f_{i,h}} (S_{f_{i,h}} z_{i,h} - \sigma_{i,h} \hat{W}_{f_{i,h}}) \\ \dot{\hat{W}}_{c_{i,h}} &= -\gamma_{c_{i,h}} S_{\mathcal{J}_{i,h}} S_{\mathcal{J}_{i,h}}^\top \hat{W}_{c_{i,h}} \\ \dot{\hat{W}}_{ai,h} &= \dot{\hat{W}}_{c_{i,h}} - \gamma_{ai,h} S_{\mathcal{J}_{i,h}} S_{\mathcal{J}_{i,h}}^\top (\hat{W}_{ai,h} - \hat{W}_{c_{i,h}}).\end{aligned}\quad (45)$$

Thus, the sub-actor-critic-identifier framework of the $z_{i,h}$ sub-system is presented as follows:

$$\begin{aligned}\nabla \hat{\mathcal{J}}_{i,h}^* &= 2 \left(k_{i,h}\Theta(z_{i,h}) + \hat{W}_{f_{i,h}}^\top S_{f_{i,h}} + \frac{5}{2}z_{i,h} \right. \\ &\quad \left. + \Xi_{i,h} - \dot{\alpha}_{if,h} + \hat{W}_{c_{i,h}}^\top S_{\mathcal{J}_{i,h}} \right), \\ \hat{\alpha}_{i,h}^* &= - \left(k_{i,h}\Theta(z_{i,h}) + \hat{W}_{f_{i,h}}^\top S_{f_{i,h}} + \frac{5}{2}z_{i,h} \right. \\ &\quad \left. + \Xi_{i,h} - \dot{\alpha}_{if,h} + \hat{W}_{ai,h}^\top S_{\mathcal{J}_{i,h}} \right).\end{aligned}\quad (46)$$

Through the weight updates of the NN, the HJB equation ultimately evolves into

$$\begin{aligned}\hat{\mathcal{H}}_{i,h} \left(z_{i,h}, \hat{\alpha}_{i,h}^*, \nabla \hat{\mathcal{J}}_{i,h}^* \right) &= z_{i,h}^2 + \hat{\alpha}_{i,h}^{*2} + \nabla \hat{\mathcal{J}}_{i,h}^* \\ &\quad \times (\hat{\alpha}_{i,h}^* + \Upsilon_{i,h} + \Xi_{i,h} - \dot{\alpha}_{if,h}).\end{aligned}\quad (47)$$

In addition, the Bellman residual $\mathcal{H}_{i,h}$ is also provided

$$\begin{aligned}\mathcal{H}_{i,h} &= \hat{\mathcal{H}}_{i,h} \left(z_{i,h}, \hat{\alpha}_{i,h}^*, \nabla \hat{\mathcal{J}}_{i,h}^* \right) - \mathcal{H}_{i,h} \left(z_{i,h}, \alpha_{i,h}^*, \nabla \mathcal{J}_{i,h}^* \right) \\ &= \hat{\mathcal{H}}_{i,h} \left(z_{i,h}, \hat{\alpha}_{i,h}^*, \nabla \hat{\mathcal{J}}_{i,h}^* \right).\end{aligned}\quad (48)$$

Step n : Based on (15) and (16), $\dot{z}_{i,n}$ is given by

$$\dot{z}_{i,n} = u_{ia} + \Gamma_{i,n} + \Xi_{i,n} + \mathcal{F}_i - \dot{\alpha}_{if,n}. \quad (49)$$

The corresponding performance index function is also established as follows:

$$\mathcal{J}_{i,n}(z_{i,n}) = \int_t^{t_f} r_{i,n}(z_{i,n}(s), u_{ia}(z_{i,n})) ds. \quad (50)$$

Accordingly, $\mathcal{J}_{i,n}^*$ has

$$\begin{aligned}\mathcal{J}_{i,n}^* &= \min_{u_{ia} \in \mathcal{M}_n} \left(\int_t^{t_f} r_{i,n}(z_{i,n}(s), u_{ia}(z_{i,n})) ds \right) \\ &= \int_t^{t_f} r_{i,n}(z_{i,n}(s), u_{ia}^*(z_{i,n})) ds\end{aligned}\quad (51)$$

where u_{ia}^* stands for the optimal controller for actuator faults and $\mathcal{M}_n$ represents the admissible strategy sets.

In light of (51), the HJB equation is derived as follows:

$$\begin{aligned}\mathcal{H}_{i,n}(z_{i,n}, u_{ia}^*, \nabla \mathcal{J}_{i,n}^*) &= r_{i,n}(z_{i,n}, u_{ia}^*) + \nabla \mathcal{J}_{i,n}^* \dot{z}_{i,n} \\ &= z_{i,n}^2 + u_{ia}^{*2} + \nabla \mathcal{J}_{i,n}^* \\ &\quad \times (u_{ia}^* + \Gamma_{i,n} + \Xi_{i,n} + \mathcal{F}_i - \dot{\alpha}_{if,n}) = 0.\end{aligned}\quad (52)$$

Since $\mathcal{H}(z_{i,n}, u_{ia}^*, \nabla \mathcal{J}_{i,n}^*) = 0$ and $\partial \mathcal{H} / \partial u_{ia}^* = 0$, the optimal controller u_{ia}^* is

$$\frac{\partial \mathcal{H}_{i,n}}{\partial u_{ia}^*} = 0 \Rightarrow u_{ia}^* = -\frac{1}{2} \nabla \mathcal{J}_{i,n}^*. \quad (53)$$

Repeating the previous steps, $\nabla \mathcal{J}_{i,n}^*$ and u_{ia}^* are split into

$$\begin{aligned}\nabla \mathcal{J}_{i,n}^* &= 2 \left(k_{i,n}\Theta(z_{i,n}) + \Upsilon_{i,n} + \Xi_{i,n} \right. \\ &\quad \left. + \frac{3}{2}z_{i,n} - \dot{\alpha}_{if,n} + \nabla \tilde{\mathcal{J}}_{i,n}^* \right) \\ u_{ia}^* &= - \left(k_{i,n}\Theta(z_{i,n}) + \Upsilon_{i,n} + \Xi_{i,n} \right. \\ &\quad \left. + \frac{3}{2}z_{i,n} - \dot{\alpha}_{if,n} + \nabla \tilde{\mathcal{J}}_{i,n}^* \right)\end{aligned}\quad (54)$$

where $\nabla \tilde{\mathcal{J}}_{i,n}^* = -k_{i,n}\Theta(z_{i,n}) - \Upsilon_{i,n} - \Xi_{i,n} - (3/2)z_{i,n} + \dot{\alpha}_{if,n} + (1/2)\nabla \mathcal{J}_{i,n}^* \in \mathbb{R}$ and $\Upsilon_{i,n} = \Gamma_{i,n} + \mathcal{F}_i$ are the unknown continuous function, $\Theta_{i,n}(z_{i,n}) = \text{sig}^{2\zeta-1}(z_{i,n}) \tanh(|z_{i,n}|^{2\zeta}/\epsilon_*)$.

With the support of NNs, $\nabla \tilde{\mathcal{J}}_{i,n}^*$ and $\Upsilon_{i,n}$ are approximated as follows:

$$\begin{aligned}\Upsilon_{i,n} &= W_{f_{i,n}}^{*\top} S_{f_{i,n}} + \varepsilon_{f_{i,n}} \\ \nabla \tilde{\mathcal{J}}_{i,n}^* &= W_{\mathcal{J}_{i,n}}^{*\top} S_{\mathcal{J}_{i,n}} + \varepsilon_{\mathcal{J}_{i,n}}\end{aligned}\quad (55)$$

where $W_{f_{i,n}}^*$ and $W_{\mathcal{J}_{i,n}}^*$ denote the ideal weights, $S_{f_{i,n}}$ and $S_{\mathcal{J}_{i,n}}$ stand for the activation functions, and $\varepsilon_{f_{i,n}}$ and $\varepsilon_{\mathcal{J}_{i,n}}$ are the approximation errors.

Based on (55), (54) changes into

$$\nabla \mathcal{J}_{i,n}^* = 2 \left(k_{i,n}\Theta(z_{i,n}) + W_{f_{i,n}}^{*\top} S_{f_{i,n}} + \Xi_{i,n} \right.$$

$$u_{ia}^* = - \left(k_{i,n} \Theta(z_{i,n}) + W_{f_{i,n}}^* S_{f_{i,n}} + \Xi_{i,n} + \frac{3}{2} z_{i,n} - \dot{\alpha}_{if,n} + W_{f_{i,n}}^* S_{f_{i,n}} + \varepsilon_{i,n} \right) \quad (56)$$

where $\varepsilon_{i,n} = \varepsilon_{f_{i,n}} + \varepsilon_{J_{i,n}}$.

Then, the local actor-critic-identifier NN is established as follows:

$$\begin{aligned} \Upsilon_{i,n} &\Leftarrow \hat{\Upsilon}_{i,n} = \hat{W}_{f_{i,n}}^T S_{f_{i,n}} \\ \nabla \hat{J}_{i,n}^* &\Leftarrow \nabla \hat{J}_{c_{i,n}} = \hat{W}_{c_{i,n}}^T S_{J_{i,n}} \\ \nabla \hat{J}_{i,n}^* &\Leftarrow \nabla \hat{J}_{a_{i,n}} = \hat{W}_{a_{i,n}}^T S_{J_{i,n}} \end{aligned} \quad (57)$$

where $\hat{W}_{f_{i,n}}$, $\hat{W}_{c_{i,n}}$, and $\hat{W}_{a_{i,n}}$ are weight estimates of $W_{f_{i,n}}^*$, $W_{c_{i,n}}^*$, and $W_{a_{i,n}}^*$, respectively. The corresponding update rate of NNs is designed as follows:

$$\begin{aligned} \dot{\hat{W}}_{f_{i,n}} &= \gamma_{f_{i,n}} (S_{f_{i,n}} z_{i,n} - \sigma_{i,n} \hat{W}_{f_{i,n}}) \\ \dot{\hat{W}}_{c_{i,n}} &= -\gamma_{c_{i,n}} S_{J_{i,n}} S_{J_{i,n}}^T \hat{W}_{c_{i,n}} \\ \dot{\hat{W}}_{a_{i,n}} &= \hat{W}_{c_{i,n}} - \gamma_{a_{i,n}} S_{J_{i,n}} S_{J_{i,n}}^T (\hat{W}_{a_{i,n}} - \hat{W}_{c_{i,n}}) \end{aligned} \quad (58)$$

where $\gamma_{f_{i,n}}, \gamma_{c_{i,n}}, \gamma_{a_{i,n}} > 0$ are learning rates and $\sigma_i > 0$ is a adjustment parameter.

Ultimately, the sub-actor-critic-identifier framework of the $z_{i,n}$ subsystem is constructed as follows:

$$\begin{aligned} \nabla \hat{J}_{i,n}^* &= 2 \left(k_{i,n} \Theta(z_{i,n}) + \hat{W}_{f_{i,n}}^T S_{f_{i,n}} + \Xi_{i,n} + \frac{3}{2} z_{i,n} - \dot{\alpha}_{if,n} + \hat{W}_{c_{i,n}}^T S_{J_{i,n}} \right), \\ \hat{u}_{ia}^* &= - \left(k_{i,n} \Theta(z_{i,n}) + \hat{W}_{f_{i,n}}^T S_{f_{i,n}} + \Xi_{i,n} + \frac{3}{2} z_{i,n} - \dot{\alpha}_{if,n} + \hat{W}_{a_{i,n}}^T S_{J_{i,n}} \right). \end{aligned} \quad (59)$$

It is obvious that the HJB equation can be approximated as follows:

$$\begin{aligned} \hat{\mathcal{H}}_{i,n}(z_{i,n}, \hat{u}_{ia}^*, \nabla \hat{J}_{i,n}^*) &= r_{i,n}(z_{i,n}, \hat{u}_{ia}^*) + \nabla \hat{J}_{i,n}^* \dot{z}_{i,n} \\ &= z_{i,n}^2 + \hat{u}_{ia}^{*2} + \nabla \hat{J}_{i,n}^* \\ &\quad \times (\hat{u}_{ia} + \Upsilon_{i,h} + \Xi_{i,n} - \dot{\alpha}_{if,n}). \end{aligned} \quad (60)$$

Combining (52) and (60), the Bellman residual $\mathcal{H}_{i,n}$ is obtained

$$\begin{aligned} \mathcal{H}_{i,n} &= \hat{\mathcal{H}}_{i,n}(z_{i,n}, \hat{u}_{ia}^*, \nabla \hat{J}_{i,n}^*) - \mathcal{H}_{i,n}(z_{i,n}, u_{ia}^*, \nabla J_{i,n}^*) \\ &= \hat{\mathcal{H}}_{i,n}(z_{i,n}, \hat{u}_{ia}^*, \nabla \hat{J}_{i,n}^*). \end{aligned} \quad (61)$$

Within the sub-actor-critic-identifier framework, the controllers are optimally designed based on identification, evaluation, and execution information to achieve optimal control performance. Notably, each virtual controller is specifically tailored to provide an optimized solution for its corresponding subsystem, thereby enhancing the overall system performance with respect to safe and control precision.

E. Stability Analysis

Theorem 1: Considering that the MASs composed of leaders and followers is plagued by safe concerns, under the guarantees provided by Assumptions 1 and 2 and Lemmas 1–5, and with updates from the sub-actor-critic-identifier framework, if the optimal controller (59) is used and the parameter conditions are satisfied, then the following conditions hold.

- 1) All system states can satisfy the constraint $-\underline{\rho}_{i,h} < x_{i,h} < \bar{\rho}_{i,h}$ after a specified time T_d .
- 2) The closed-loop MASs maintain stability while guaranteeing boundedness of all system variables.
- 3) The outputs of follower agents are confined within the convex hull generated by the leader states.

Proof: Define $\tilde{W}_{f_{i,h}} = \hat{W}_{f_{i,h}} - W_{f_{i,h}}^*$, $\tilde{W}_{a_{i,h}} = \hat{W}_{a_{i,h}} - W_{a_{i,h}}^*$, $\tilde{W}_{c_{i,h}} = \hat{W}_{c_{i,h}} - W_{c_{i,h}}^*$, $h = 1, \dots, n$ as the estimation errors of the weight vectors.

Step 1: For $z_{i,1}$ subsystem, choose the following candidate Lyapunov function:

$$\begin{aligned} \mathcal{V}_{i,1} &= \frac{1}{2} z_{i,1}^2 + \frac{1}{2} \tilde{W}_{f_{i,1}}^T \gamma_{f_{i,1}}^{-1} \tilde{W}_{f_{i,1}} + \frac{1}{2} \tilde{W}_{a_{i,1}}^T \tilde{W}_{a_{i,1}} \\ &\quad + \frac{1}{2} \tilde{W}_{c_{i,1}}^T \tilde{W}_{c_{i,1}} + \frac{1}{2} \mathcal{B}_{i,2}^2. \end{aligned} \quad (62)$$

Computing the time derivative of $\mathcal{V}_{i,1}$, substitute (16) and (27)–(30) into it and and simplify

$$\begin{aligned} \dot{\mathcal{V}}_{i,1} &= -k_{i,1} z_{i,1} \Theta(z_{i,1}) + \mathcal{E}_i z_{i,1} z_{i,2} + z_{i,1} \varepsilon_{i,1} + \mathcal{E}_i z_{i,1} \mathcal{B}_{i,2} \\ &\quad - (1 + \mathcal{E}_i^2) z_{i,1} - z_{i,1} \hat{W}_{a_{i,1}}^T S_{J_{i,1}} - \sigma_{i,1} \tilde{W}_{f_{i,1}}^T \hat{W}_{f_{i,1}} \\ &\quad + (\gamma_{a_{i,1}} - \gamma_{c_{i,1}}) \tilde{W}_{a_{i,1}}^T S_{J_{i,1}} S_{J_{i,1}}^T \hat{W}_{c_{i,1}} \\ &\quad - \gamma_{a_{i,1}} \tilde{W}_{a_{i,1}}^T S_{J_{i,1}} S_{J_{i,1}}^T \hat{W}_{a_{i,1}} - \mathcal{B}_{i,2} \dot{\alpha}_{i,1}^* \\ &\quad - \gamma_{c_{i,1}} \tilde{W}_{c_{i,1}}^T S_{J_{i,1}} S_{J_{i,1}}^T \hat{W}_{c_{i,1}} - \frac{\mathcal{B}_{i,2}^2}{\tau_{i,2}} \end{aligned} \quad (63)$$

where $\dot{\alpha}_{i,1}^* = (\partial \hat{\alpha}_{i,1}^* / \partial z_{i,1}) \dot{z}_{i,1} + (\partial \hat{\alpha}_{i,1}^* / \partial \dot{z}_{i,1}) \ddot{z}_{i,1} + (\partial \hat{\alpha}_{i,1}^* / \partial \hat{W}_{a_{i,1}}) \dot{\hat{W}}_{a_{i,1}} + (\partial \hat{\alpha}_{i,1}^* / \partial \hat{W}_{c_{i,1}}) \dot{\hat{W}}_{c_{i,1}} + (\partial \hat{\alpha}_{i,1}^* / \partial \hat{W}_{f_{i,1}}) \dot{\hat{W}}_{f_{i,1}}$ is a continuous function and $|\dot{\alpha}_{i,1}^*| \leq \Pi_{i,1}$.

Then, by using Young's inequality, we have

$$\begin{aligned} \mathcal{E}_i z_{i,1} z_{i,2} &\leq \frac{1}{2} \mathcal{E}_i^2 z_{i,1}^2 + \frac{1}{2} z_{i,2}^2, \\ z_{i,1} \varepsilon_{i,1} &\leq \frac{1}{2} z_{i,1}^2 + \frac{1}{2} \varepsilon_{i,1}^2, \\ \mathcal{E}_i z_{i,1} \mathcal{B}_{i,2} &\leq \frac{1}{2} \mathcal{E}_i^2 z_{i,1}^2 + \frac{1}{2} \mathcal{B}_{i,2}^2, \\ -\mathcal{B}_{i,2} \dot{\alpha}_{i,1}^* &\leq \frac{1}{2} \mathcal{B}_{i,2}^2 + \frac{1}{2} \Pi_{i,1}^2, \\ -z_{i,1} \hat{W}_{a_{i,1}}^T S_{J_{i,1}} &\leq \frac{1}{2} z_{i,1}^2 + \frac{1}{2} (\hat{W}_{a_{i,1}}^T S_{J_{i,1}})^2, \\ -\sigma_{i,1} \tilde{W}_{f_{i,1}}^T \hat{W}_{f_{i,1}} &\leq -\frac{\sigma_{i,1}}{2} \tilde{W}_{f_{i,1}}^T \tilde{W}_{f_{i,1}} + \frac{\sigma_{i,1}}{2} W_{f_{i,1}}^* W_{f_{i,1}}^*, \\ \tilde{W}_{a_{i,1}}^T S_{J_{i,1}} S_{J_{i,1}}^T \hat{W}_{c_{i,1}} &\leq \frac{1}{2} (\tilde{W}_{a_{i,1}}^T S_{J_{i,1}})^2 + \frac{1}{2} (\hat{W}_{c_{i,1}}^T S_{J_{i,1}})^2. \end{aligned} \quad (64)$$

Furthermore

$$\tilde{W}_{a_{i,1}}^T S_{J_{i,1}} S_{J_{i,1}}^T \hat{W}_{a_{i,1}} = \frac{1}{2} (\tilde{W}_{a_{i,1}}^T S_{J_{i,1}})^2 + \frac{1}{2} (\hat{W}_{a_{i,1}}^T S_{J_{i,1}})^2$$

$$\begin{aligned} & -\frac{1}{2} \left(\tilde{W}_{a_{i,1}}^* S_{\mathcal{J}_{i,1}} \right)^2, \\ \tilde{W}_{c_{i,1}}^T S_{\mathcal{J}_{i,1}} S_{\mathcal{J}_{i,1}}^T \hat{W}_{c_{i,1}} &= \frac{1}{2} \left(\tilde{W}_{c_{i,1}}^T S_{\mathcal{J}_{i,1}} \right)^2 + \frac{1}{2} \left(\hat{W}_{c_{i,1}}^T S_{\mathcal{J}_{i,1}} \right)^2 \\ & -\frac{1}{2} \left(\tilde{W}_{c_{i,1}}^* S_{\mathcal{J}_{i,1}} \right)^2. \end{aligned} \quad (65)$$

Therefore, $\dot{V}_{i,1}$ is presented as follows:

$$\begin{aligned} \dot{V}_{i,1} &\leq \frac{1}{2} z_{i,2}^2 - k_{i,1} z_{i,1}^{2\zeta} - \frac{\sigma_{i,1}}{2} \tilde{W}_{f_{i,1}}^T \tilde{W}_{f_{i,1}} \\ &\quad - \gamma_{c_{i,1}} \lambda_{\min}(S_{\mathcal{J}_{i,1}}) \tilde{W}_{a_{i,1}}^T \tilde{W}_{a_{i,1}} - \chi_{i,1} \mathcal{B}_{i,2}^2 \\ &\quad - \gamma_{c_{i,1}} \lambda_{\min}(S_{\mathcal{J}_{i,1}}) \tilde{W}_{c_{i,1}}^T \tilde{W}_{c_{i,1}} + \Phi_1 \end{aligned} \quad (66)$$

where $\Phi_1 = (1/2)(\tilde{\varepsilon}_{f_{i,1}}^2 + \Pi_{i,1}^2 + \sigma_{i,1} W_{f_{i,1}}^* W_{f_{i,1}}^* + \gamma_{a_{i,1}} \lambda_{\min}(S_{\mathcal{J}_{i,1}}) W_{a_{i,1}}^* W_{a_{i,1}}^* + \gamma_{c_{i,1}} \lambda_{\min}(S_{\mathcal{J}_{i,1}}) W_{c_{i,1}}^* W_{c_{i,1}}^*) + k_{i,1} \varsigma \epsilon_*$ and $\chi_{i,2} = ((1/\tau_{i,2}) - 1) > 0$, $1 < \gamma_{a_{i,1}} < 2\gamma_{c_{i,1}}$.

Step h: Select a candidate Lyapunov function with respect to the $z_{i,h}$ subsystem

$$\begin{aligned} V_{i,h} &= \frac{1}{2} z_{i,h}^2 + \frac{1}{2} \tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}} + \frac{1}{2} \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} \\ &\quad + \frac{1}{2} \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} + \frac{1}{2} \mathcal{B}_{i,h+1}^2. \end{aligned} \quad (67)$$

$\dot{V}_{i,h}$ can be organized as follows:

$$\begin{aligned} \dot{V}_{i,h} &= -k_{i,h} z_{i,h} \Theta(z_{i,h}) + z_{i,h} z_{i,h+1} + z_{i,h} \varepsilon_{i,h} + z_{i,h} \mathcal{B}_{i,h+1} \\ &\quad - \frac{5}{2} z_{i,h}^2 - z_{i,h} \hat{W}_{a_{i,h}}^T S_{\mathcal{J}_{i,h}} - \sigma_{i,h} \tilde{W}_{f_{i,h}}^T \tilde{W}_{f_{i,h}} \\ &\quad + (\gamma_{a_{i,h}} - \gamma_{c_{i,h}}) \tilde{W}_{a_{i,h}}^T S_{\mathcal{J}_{i,h}} S_{\mathcal{J}_{i,h}}^T \hat{W}_{c_{i,h}} \\ &\quad - \gamma_{a_{i,h}} \tilde{W}_{a_{i,h}}^T S_{\mathcal{J}_{i,h}} S_{\mathcal{J}_{i,h}}^T \hat{W}_{a_{i,h}} - \mathcal{B}_{i,h+1} \dot{\hat{\alpha}}_{i,h}^* \\ &\quad - \gamma_{c_{i,h}} \tilde{W}_{c_{i,h}}^T S_{\mathcal{J}_{i,h}} S_{\mathcal{J}_{i,h}}^T \hat{W}_{c_{i,h}} - \frac{\mathcal{B}_{i,h+1}^2}{\tau_{i,h+1}} \end{aligned} \quad (68)$$

where $\dot{\hat{\alpha}}_{i,h}^* = (\partial \hat{\alpha}_{i,1}^* / \partial z_{i,h}) \dot{z}_{i,h} + (\hat{\alpha}_{i,h}^* / \partial \hat{W}_{a_{i,h}}) \dot{\hat{W}}_{a_{i,h}} + (\hat{\alpha}_{i,h}^* / \partial \hat{W}_{c_{i,h}}) \dot{\hat{W}}_{c_{i,h}} + (\hat{\alpha}_{i,h}^* / \partial \hat{W}_{f_{i,h}}) \dot{\hat{W}}_{f_{i,h}}$ is a continuous function and $|\dot{\hat{\alpha}}_{i,h}^*| \leq \Pi_h$. Ultimately, the result of $\dot{V}_{i,h}$ has

$$\begin{aligned} \dot{V}_{i,h} &\leq \frac{1}{2} z_{i,h+1}^2 - k_{i,h} z_{i,h}^{2\zeta} - \frac{\sigma_{i,h}}{2} \tilde{W}_{f_{i,h}}^T \tilde{W}_{f_{i,h}} \\ &\quad - \frac{\gamma_{c_{i,h}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} - \chi_{i,h+1} \mathcal{B}_{i,h+1}^2 \\ &\quad - \frac{\gamma_{c_{i,h}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} + \Phi_h \end{aligned} \quad (69)$$

where $\Phi_h = (1/2)(\tilde{\varepsilon}_{f_{i,h}}^2 + \Pi_{i,h}^2 + \sigma_{i,h} W_{f_{i,h}}^* W_{f_{i,h}}^* + \gamma_{a_{i,h}} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) W_{a_{i,h}}^* W_{a_{i,h}}^* + \gamma_{c_{i,h}} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) W_{c_{i,h}}^* W_{c_{i,h}}^*) + k_{i,h} \varsigma \epsilon_*$ and $\chi_{i,h+1} = ((1/\tau_{i,h+1}) - 1) > 0$, $1 < \gamma_{a_{i,h}} < 2\gamma_{c_{i,h}}$.

Step n: Different from the previous steps, the candidate Lyapunov function has a minor adjustment

$$\begin{aligned} V_{i,n} &= \frac{1}{2} z_{i,n}^2 + \frac{1}{2} \tilde{W}_{f_{i,n}}^T \gamma_{f_{i,n}}^{-1} \tilde{W}_{f_{i,n}} + \frac{1}{2} \tilde{W}_{a_{i,n}}^T \tilde{W}_{a_{i,n}} \\ &\quad + \frac{1}{2} \tilde{W}_{c_{i,n}}^T \tilde{W}_{c_{i,n}}. \end{aligned} \quad (70)$$

$\dot{V}_{i,n}$ is derived as follows:

$$\begin{aligned} \dot{V}_{i,n} &= -k_{i,n} z_{i,n} \Theta(z_{i,n}) + z_{i,n} \varepsilon_{i,n} - \frac{3}{2} z_{i,n}^2 - z_{i,n} \hat{W}_{a_{i,n}}^T S_{\mathcal{J}_{i,n}} \\ &\quad - \sigma_{i,n} \tilde{W}_{f_{i,n}}^T \tilde{W}_{f_{i,n}} - \gamma_{c_{i,n}} \tilde{W}_{c_{i,n}}^T S_{\mathcal{J}_{i,n}} S_{\mathcal{J}_{i,n}}^T \hat{W}_{c_{i,n}} \end{aligned}$$

$$\begin{aligned} &+ (\gamma_{a_{i,n}} - \gamma_{c_{i,n}}) \tilde{W}_{a_{i,n}}^T S_{\mathcal{J}_{i,n}} S_{\mathcal{J}_{i,n}}^T \hat{W}_{c_{i,n}} \\ &- \gamma_{a_{i,n}} \tilde{W}_{a_{i,n}}^T S_{\mathcal{J}_{i,n}} S_{\mathcal{J}_{i,n}}^T \hat{W}_{a_{i,n}}. \end{aligned} \quad (71)$$

Similar to *Step 1* and *Step h*, $\dot{V}_{i,n}$ has

$$\begin{aligned} \dot{V}_{i,n} &\leq -k_{i,n} z_{i,n}^{2\zeta} - \frac{\sigma_{i,n}}{2} \tilde{W}_{f_{i,n}}^T \tilde{W}_{f_{i,n}} \\ &\quad - \frac{\gamma_{c_{i,n}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,n}}) \tilde{W}_{a_{i,n}}^T \tilde{W}_{a_{i,n}} \\ &\quad - \frac{\gamma_{c_{i,n}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,n}}) \tilde{W}_{c_{i,n}}^T \tilde{W}_{c_{i,n}} + \Phi_n. \end{aligned} \quad (72)$$

where $\Phi_n = k_{i,n} \varsigma \epsilon_* + (1/2)(\tilde{\varepsilon}_{f_{i,n}}^2 + \sigma_{i,n} W_{f_{i,n}}^* W_{f_{i,n}}^* + \gamma_{a_{i,n}} \lambda_{\min}(S_{\mathcal{J}_{i,n}}) W_{a_{i,n}}^* W_{a_{i,n}}^* + \gamma_{c_{i,n}} \lambda_{\min}(S_{\mathcal{J}_{i,n}}) W_{c_{i,n}}^* W_{c_{i,n}}^*)$ and $1 < \gamma_{a_{i,h}} < 2\gamma_{c_{i,h}}$.

Choose candidate Lyapunov functions for the entire system

$$\begin{aligned} V &= \sum_{i=1}^N \sum_{h=1}^n \left(\frac{1}{2} z_{i,h}^2 + \frac{1}{2} \tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}} + \frac{1}{2} \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} \right. \\ &\quad \left. + \frac{1}{2} \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} \right) + \sum_{i=1}^N \sum_{h=2}^n \frac{1}{2} \mathcal{B}_{i,h}^2. \end{aligned} \quad (73)$$

Based on the above information, $\dot{V}$ is summarized as follows:

$$\begin{aligned} \dot{V} &\leq \sum_{i=1}^N \sum_{h=1}^n \left(-k_{i,h} z_{i,h}^{2\zeta} - \frac{\gamma_{c_{i,h}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} \right. \\ &\quad \left. - \frac{\sigma_{i,h}}{2} \tilde{W}_{f_{i,h}}^T \tilde{W}_{f_{i,h}} - \frac{\gamma_{c_{i,h}}}{2} \lambda_{\min}(S_{\mathcal{J}_{i,h}}) \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} \right) \\ &\quad - \sum_{i=1}^N \sum_{h=2}^n \chi_{i,h} \mathcal{B}_{i,h}^2 + \sum_{h=1}^n \Phi_h \\ &= \sum_{i=1}^N \sum_{h=1}^n \left(-\rho_{i1,h} z_{i,h}^{2\zeta} - \rho_{i2,h} \tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}} \right. \\ &\quad \left. - \rho_{i3,h} \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} - \rho_{i4,h} \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} \right) \\ &\quad - \sum_{i=1}^N \sum_{h=2}^n \rho_{i5,h} \mathcal{B}_{i,h}^2 + \sum_{h=1}^n \Phi_h \end{aligned} \quad (74)$$

where $\rho_{i1,h} = k_{i,h}$, $\rho_{i2,h} = (\sigma_{i,h}/2)$, $\rho_{i3,h} = (\gamma_{c_{i,h}}/2) \lambda_{\min}(S_{\mathcal{J}_{i,h}})$, $\rho_{i4,h} = (\gamma_{c_{i,h}}/2) \lambda_{\min}(S_{\mathcal{J}_{i,h}})$, and $\rho_{i5,h} = \chi_{i,h}$. By using Lemma 5, it has

$$\begin{aligned} \left(\tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}} \right)^{\zeta} &\leq (1 - \varsigma) \varsigma^{\frac{\zeta}{1-\varsigma}} + \tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}}, \\ \left(\tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} \right)^{\zeta} &\leq (1 - \varsigma) \varsigma^{\frac{\zeta}{1-\varsigma}} + \tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}}, \\ \left(\tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} \right)^{\zeta} &\leq (1 - \varsigma) \varsigma^{\frac{\zeta}{1-\varsigma}} + \tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}}, \\ \mathcal{B}_{i,h}^{2\zeta} &\leq (1 - \varsigma) \varsigma^{\frac{\zeta}{1-\varsigma}} + \mathcal{B}_{i,h}^2. \end{aligned} \quad (75)$$

Based on Lemma 4, $\dot{V}$ is further derived as follows:

$$\begin{aligned} \dot{V} &\leq \sum_{i=1}^N \sum_{h=1}^n \left(-\rho_{i1,p} \left(z_{i,h}^2 \right)^{\zeta} - \rho_{i2,h} \left(\tilde{W}_{f_{i,h}}^T \gamma_{f_{i,h}}^{-1} \tilde{W}_{f_{i,h}} \right)^{\zeta} \right. \\ &\quad \left. - \rho_{i3,h} \left(\tilde{W}_{a_{i,h}}^T \tilde{W}_{a_{i,h}} \right)^{\zeta} - \rho_{i4,h} \left(\tilde{W}_{c_{i,h}}^T \tilde{W}_{c_{i,h}} \right)^{\zeta} \right) \\ &\quad - \sum_{i=1}^N \sum_{h=2}^n \rho_{i5,p} \left(\mathcal{B}_{i,h}^2 \right)^{\zeta} + \Phi \\ &\leq -\rho V^{\zeta} + \Phi \end{aligned} \quad (76)$$

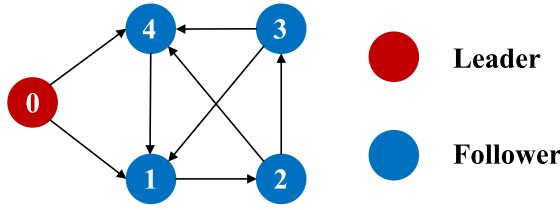


Fig. 1. Communication topology.

where $\rho = \min\{\min\{2^\zeta \rho_{i1,h}\}, \min\{2^\zeta \rho_{i2,h}\}, \min\{2^\zeta \rho_{i3,h}\}, \min\{2^\zeta \rho_{i4,h}\}, \min\{2^\zeta \rho_{i5,h}\}\}$. $\Phi = (1 - \zeta)\zeta^{(\zeta/1-\zeta)}(\sum_{i=1}^N \sum_{h=1}^n (\rho_{i2,h} + \rho_{i3,h} + \rho_{i4,h}) + \sum_{i=1}^N \sum_{h=1}^{n-1} \rho_{i5,h}) + \sum_{h=1}^n \Phi_h$. It can be drawn a conclusion from (76) that $\mathcal{V} \in \mathcal{L}_\infty$ for any initial conditions, which implies that $z_{i,h} \in \mathcal{L}_\infty$, $\tilde{W}_{f_{i,h}} \in \mathcal{L}_\infty$, $\tilde{W}_{a_{i,h}} \in \mathcal{L}_\infty$, $\tilde{W}_{c_{i,h}} \in \mathcal{L}_\infty$, and $\mathcal{B}_{i,h} \in \mathcal{L}_\infty$. Therefore, the entire closed-loop MASs keep stable; meanwhile, the all signals remaining bounded and $\|z_{i,h}\| \leq (2((\Phi/((1-\rho)\rho)))^{1/5})^{1/2}$. The control objectives and lemmas are successfully realized.

Remark 7: It should be emphasized that this study primarily employs a policy iteration algorithm. Selecting an appropriate initial policy is critical for MASs stability, as it ensures the proper execution of policy iteration and yields optimized controllers. However, the OB technique inherently guarantees subsystem stability through its focus on optimizing each subsystem within the MASs. Furthermore, the incorporation of terminal time constraints enhances overall system stability and robustness. Consequently, the stringent requirement for an initial policy can be relaxed to a certain extent.

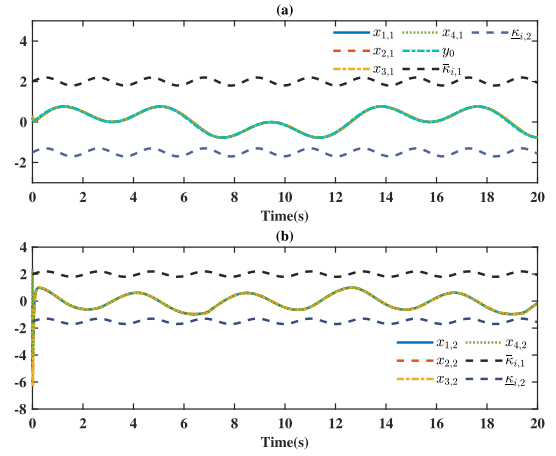
Remark 8: Owing to the estimation errors $\tilde{W}_{c_{i,h}}$ and $\tilde{W}_{a_{i,h}}$ induced by the actor-critic update laws designed under the simplified RL strategy, the system is unable to satisfy the stability condition required for convergence in Lemma 5. To address this issue, Lemma 5 is further employed in this article to derive the key inequality (75), through which both stability and optimality are simultaneously guaranteed. Moreover, this approach establishes a novel adaptive estimation analysis framework that provides a rigorous alternative for handling the inherent estimation errors associated with NN learning.

IV. SIMULATION RESULTS

To verify the effectiveness of the proposed method, a simulation example is presented in this section. The nonlinear MASs interconnected via topology graph (Fig. 1), and the model of each follower is

$$\begin{cases} \dot{x}_{i,1} = x_{i,2} + f_{i,1}(x_{i,1}) \\ \dot{x}_{i,2} = u_i^F + f_{i,2}(\bar{x}_{i,2}) \\ y_i = x_{i,1}, i = 1, \dots, 4 \end{cases} \quad (77)$$

where $f_{i,1}(x_{i,1}) = 0.1x_{i,1}^2$, $f_{i,2}(\bar{x}_{i,2}) = 0.1x_{i,1}x_{i,2}^2$ and leader is seen as $y_0 = \sin(t) \cos(0.5t)$. In this example, we assume that agent 2 experiences a drift fault, agent 3 experiences a loss fault, agent 4 experiences a bias fault, and agent 1 remains


 Fig. 2. (a) Responses of the state $x_{i,1}$ in the constrained MASs. (b) Responses of the state $x_{i,2}$ in the constrained MASs.

fault-free

$$\begin{aligned} u_2^F(t) &= \begin{cases} u_{2a} + 0.5t, & 8 \leq t \leq 20 \\ u_{2a}, & \text{otherwise} \end{cases} \\ u_3^F(t) &= \begin{cases} (0.3 - 0.1 \sin(t)) u_{3a}, & 15 \leq t \leq 20 \\ u_{3a}, & \text{otherwise} \end{cases} \\ u_4^F(t) &= \begin{cases} u_{4a} + 20, & 12 \leq t \leq 20 \\ u_{4a}, & \text{otherwise.} \end{cases} \end{aligned} \quad (78)$$

The initial values of system states are chosen as $x_{1,1}(0) = 0.2, x_{1,2}(0) = -0.3, x_{2,1}(0) = 0.25, x_{2,2}(0) = -0.4, x_{3,1}(0) = 0.3, x_{3,2}(0) = -0.5, x_{4,1}(0) = 0.15$, and $x_{4,2}(0) = 0.2, \alpha_{if}(0) = 0$. The boundary functions are set as $\bar{\rho}_{i,1} = 2 + 0.2 \sin(3t), \bar{\rho}_{i,2} = 1.5 - 0.2 \sin(3t)$. Controller parameters are given as $T_d = 3, k_{i,1} = 100, k_{i,2} = 200, \zeta = 4/5, \epsilon_* = 0.08, \tau_i = 0.0001, \gamma_{f_{i,1}} = 2, \sigma_{i,1} = 1$, and $\gamma_{c_{i,1}} = \gamma_{a_{i,1}} = 200$. The initial weights of the identifier NN are $\hat{W}_{f_{i,1}}(0) = \hat{W}_{f_{i,2}}(0) = [0.3, \dots, 0.3]^T \in \mathbb{R}^{11 \times 1}$. The initial weights for the actor NNs of the each agent are selected as $\hat{W}_{a_{1,1}}(0) = \hat{W}_{a_{1,2}}(0) = [0.4, \dots, 0.4]^T \in \mathbb{R}^{11 \times 1}$, $\hat{W}_{a_{2,1}}(0) = \hat{W}_{a_{2,2}}(0) = [0.3, \dots, 0.3]^T \in \mathbb{R}^{11 \times 1}$, $\hat{W}_{a_{3,1}}(0) = \hat{W}_{a_{3,2}}(0) = [0.2, \dots, 0.2]^T \in \mathbb{R}^{11 \times 1}$, and $\hat{W}_{a_{4,1}}(0) = \hat{W}_{a_{4,2}}(0) = [0.1, \dots, 0.1]^T \in \mathbb{R}^{11 \times 1}$. The initial weights for the critic NNs of each agent are $\hat{W}_{c_{1,1}}(0) = \hat{W}_{c_{1,2}}(0) = [0.45, \dots, 0.45]^T \in \mathbb{R}^{11 \times 1}$, $\hat{W}_{c_{2,1}}(0) = \hat{W}_{c_{2,2}}(0) = [0.35, \dots, 0.35]^T \in \mathbb{R}^{11 \times 1}$, $\hat{W}_{c_{3,1}}(0) = \hat{W}_{c_{3,2}}(0) = [0.25, \dots, 0.25]^T \in \mathbb{R}^{11 \times 1}$, and $\hat{W}_{c_{4,1}}(0) = \hat{W}_{c_{4,2}}(0) = [0.15, \dots, 0.15]^T \in \mathbb{R}^{11 \times 1}$.

Under the settings of the above parameters and initial conditions, the simulation results are shown in Figs. 2–6. As shown in Fig. 2(a), we can see that the trajectories of each follower closely track the trajectory of the leader and consistently remain within the dynamic constraints. As illustrated in Fig. 2(b), despite initial constraint violations, all agents trajectories enter and remain within the prescribed constraints after T_d , thereby confirming the delayed constraints capability to reconcile transient performance with safe guarantees. From Fig. 3, it illustrates the controller responses under different types of faults. The results show that drift faults exert the most significant impact on the controller, while loss faults have

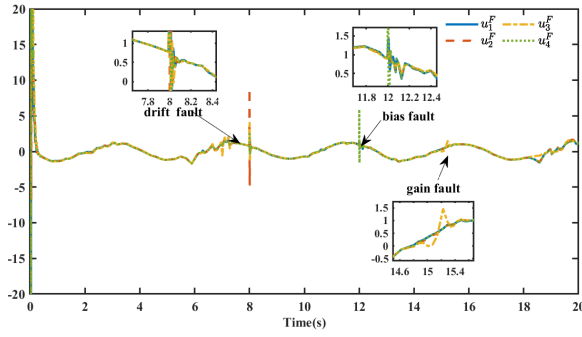


Fig. 3. Responses of the MASs optimal controller under different types of faults.

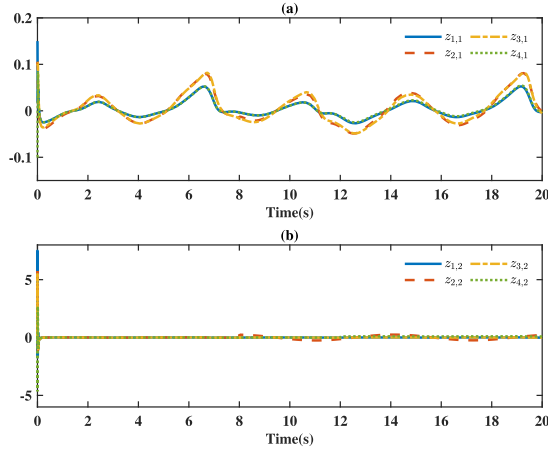


Fig. 4. (a) Responses of the state error $z_{i,1}$ in the MASs. (b) Responses of the state error $z_{i,2}$ in the MASs.

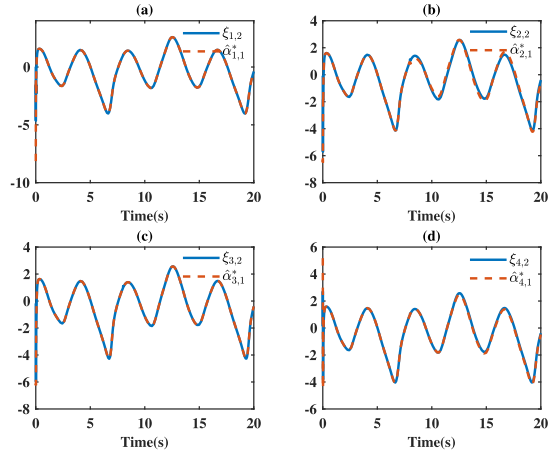


Fig. 5. (a) Responses of the optimal virtual controller $\hat{\alpha}_{1,1}^*$ and states $\xi_{1,2}$. (b) Responses of the optimal virtual controller $\hat{\alpha}_{2,1}^*$ and states $\xi_{2,2}$. (c) Responses of the optimal virtual controller $\hat{\alpha}_{3,1}^*$ and states $\xi_{3,2}$. (d) Responses of the optimal virtual controller $\hat{\alpha}_{4,1}^*$ and states $\xi_{4,2}$.

a relatively minor effect. Additionally, after a fault occurs, the controller can quickly return to a stable state due to the compensation mechanism of the identifier NN. It is also worth noting that when one agent experiences a fault, other agents are affected to varying degrees. As shown in Fig. 4, the first-order coordination errors among agents exhibit minor fluctuations around zero while maintaining overall stability.

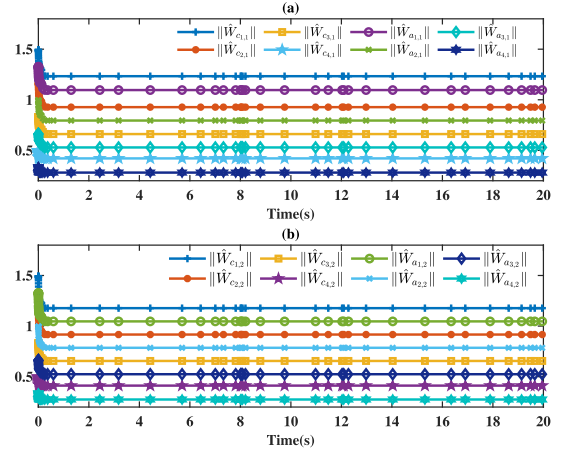


Fig. 6. (a) Response of the actor NN weights $\hat{W}_{a_{i,1}}$ and the critic NN weights $\hat{W}_{c_{i,1}}$. (b) Response of the actor NN weights $\hat{W}_{a_{i,2}}$ and the critic NN weights $\hat{W}_{c_{i,2}}$.

Except for agent 2, which is affected by an actuator fault, the second-order coordination errors of all other agents converge rapidly to zero. As illustrated in Fig. 5, each virtual controller maintains consistency with the state variables, demonstrating the system's ability to mitigate the influence of unsafe operating conditions and verifying the effectiveness of the designed safe controller. Fig. 6 illustrates that under the influence of fuzzy rules, the actor-critic NN weights converge to a stable bounded value, indicating that the RL framework enables the system to achieve near-optimal performance with minimal cost. Overall, these simulation results verify the effectiveness of the proposed optimal safe control algorithm.

V. CONCLUSION

This article proposes an optimal control strategy based on fuzzy RL to address the safe control problem of unknown nonlinear MASs. First, in the presence of coexisting safe factors such as composite constraints and multiactuator faults, the dynamic delayed state constraints and the terminal time constraints are used to provide a safe boundary and an optimal performance boundary for MASs. Meanwhile, NNs are employed to estimate the integrated nonlinear terms, including multitype fault information and unknown system information, thereby ensuring safe and stable system operation in the presence of actuator faults. Furthermore, within the framework of OB control, fuzzy RL is employed to optimize the states of each subsystem, ultimately achieving optimal performance for the entire system.

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